

Q3 CBOW

- (1) Window size
- (2) Target word
- (3) weight matrix
- (4) Hidden layer: [one hot] * weight matrix
- (5) Output layer: weight matrix * (Avg hidden)^T
- (6) Softmax: $y_j = \frac{\exp(o_j)}{\sum \exp(o_j)}$

① Window size = 2

I love to play foot ball

 \ / \ /
 cw = cw.
 ↓
 Target word.

$$W = \begin{bmatrix} 0.1 & 0.2 \\ 0.3 & 0.4 \\ 0.5 & 0.6 \\ 0.7 & 0.8 \\ 0.9 & 1 \end{bmatrix}$$

(1) One hot representation.

I [1 0 0 0 0]

love [0 1 0 0 0]

to [0 0 1 0 0]

play [0 0 0 1 0]

football [0 0 0 0 1]

(2) Hidden layer

$$I_{\text{hidden}} : [1 \ 0 \ 0 \ 0 \ 0] * W = [0.1 \ 0.2]$$

$$\text{love}_{\text{hidden}} : [0 \ 1 \ 0 \ 0 \ 0] * W = [0.3 \ 0.4]$$

$$\text{play}_{\text{hidden}} : [0 \ 0 \ 0 \ 1 \ 0] * W = [0.7 \ 0.8]$$

$$\text{football}_{\text{hidden}} : [0 \ 0 \ 0 \ 0 \ 1] * W = [0.9, 1]$$

$$\text{Avg} = [0.5, 0.6]$$

(3) Output layer

$$O = W * (\text{avg})^T = h * W^T$$

$$= \begin{bmatrix} 0.1 & 0.2 \\ 0.3 & 0.4 \\ 0.5 & 0.6 \\ 0.7 & 0.8 \\ 0.9 & 1 \end{bmatrix} * \begin{bmatrix} 0.5 \\ 0.6 \end{bmatrix}$$

$$= \begin{bmatrix} 0.05 + 0.12 \\ 0.17 \\ 0.39 \\ 0.61 \\ 0.83 + 1.05 \end{bmatrix} = \begin{bmatrix} 0.17 \\ 0.39 \\ 0.61 \\ 0.83 \\ 1.05 \end{bmatrix}$$

like to play football

(4) Soft max:

$$y_j = \frac{\exp(o_j)}{\sum_{i=1}^n \exp(o_i)}$$

o_j	$\exp(o_j)$	$y_j = \frac{\exp(o_j)}{a.649.}$
0.17	1.185	0.1223
0.39	1.477	0.153
0.61	1.8404	0.19
0.83	2.2933	0.237
1.05	2.857	0.296 ✓
Σ	<u>9.649</u>	

Highest Value is football

In iteration (1) it predicts as

football

Q4 skip gram: Predicts context words using Target word.

I love to play foot ball

↓

TW.

Step 1: One hot encoding.

I [1 0 0 0 0]

love [0 1 0 0 0]

to [0 0 1 0 0]

play [0 0 0 1 0]

football [0 0 0 0 1]

Step 2: calculate

Input Hidden layer
Only for Target word.

$$[0 \ 0 \ 1 \ 0 \ 0] * \begin{bmatrix} 0.1 & 0.2 \\ 0.3 & 0.4 \\ 0.5 & 0.6 \\ 0.7 & 0.8 \\ 0.9 & 1 \end{bmatrix}$$

$$[0.5 \ 0.6]$$

Step 3: Output Hidden layer.

$$O = h * w^T$$

$$= [0.5 \ 0.6] \begin{bmatrix} 0.1 & 0.3 & 0.5 & 0.7 & 0.9 \\ 0.2 & 0.4 & 0.6 & 0.8 & 1 \end{bmatrix}$$

$$= [0.17 \ 0.39 \ 0.61 \ 0.83 \ 1.05]$$

step 4: Soft max.

$$y_j = \frac{\exp(\sigma_j)}{\sum \exp(\sigma_j)}$$

σ_j	$\exp(\sigma_j)$	y_j
0.17	1.185	0.123
0.39	1.497	0.153
0.61	1.84	0.19
0.83	2.29	0.237
1.05	2.857	0.296
$\sum \exp \sigma_j$ 9.649		

// only difference b/w CBOW & skip gram.

↓
Content words → TW.

TW → content words.

→ use the content vectors in the hidden layer calculation.
→ use the Target vect "

② The cat sits on the mat

Window size : 2 $\downarrow$ TW.

One hot encoding.

The $[1 \ 0 \ 0 \ 0 \ 0 \ 0]$
 cat $[0 \ 1 \ 0 \ 0 \ 0 \ 0]$
 sits $[0 \ 0 \ 1 \ 0 \ 0 \ 0]$
 on $[0 \ 0 \ 0 \ 1 \ 0 \ 0]$
 the $[0 \ 0 \ 0 \ 0 \ 1 \ 0]$
 mat $[0 \ 0 \ 0 \ 0 \ 0 \ 1]$

Weights for Input to Hidden layer i .

W =
$$\begin{matrix} \text{the} \\ \text{cat} \\ \text{sits} \\ \text{on} \\ \text{mat} \\ \text{the} \\ \text{mat} \end{matrix} \begin{bmatrix} 0.1 & 0.2 & 0 & 0.5 \\ 0.3 & 0.4 & 0.1 & 0.2 \\ 0.5 & 0.3 & 0.2 & 0.6 \\ 0.2 & 0.1 & 0.3 & 0.4 \\ 0.4 & 0.5 & 0.2 & 0.1 \end{bmatrix}$$

(i) Obtain content vectors for no words surrounding "site".

[the, cat, site, on, mat]

the [1 0 0 0 0]

cat [0 1 0 0 0]

site [0 0 1 0 0]

on [0 0 0 1 0]

mat [0 0 0 0 1]

(ii) Compute Average Content Vector by adding the one hot encoded vectors of the content words and dividing by no of content words.

Avg Content vector.

$$\left[\frac{2}{4}, \frac{1}{4}, 0, \frac{1}{4}, 0 \right]$$

Input-Hidden layer:

If hidden bias is given add that value

$$[0.5 \quad 0.25 \quad 0 \quad 0.25 \quad 0]$$

$$\text{Hidden bias} : [0.1, 0.2, 0.3, 0.4]$$

$$= 0.05 + 0.075 + 0 + 0.05 + 0$$

$$= [0.625 \quad 0.275 \quad 0.425 \quad 0.4 \quad 0.5]$$

$$\begin{bmatrix} 0.1 & 0.2 & 0.0 & 0.5 \\ 0.3 & 0.4 & 0.1 & 0.2 \\ 0.5 & 0.3 & 0.2 & 0.6 \\ 0.2 & 0.1 & 0.3 & 0.4 \\ 0.4 & 0.5 & 0.2 & 0.1 \end{bmatrix}$$

∴ Hidden vector $[0.275, 0.425, 0.4, 0.5]$
 [bias added].

Output Layer

the cat sits on mat

Weight =

$$\begin{bmatrix} 0.2 & 0.1 & 0.4 & 0.5 & 0.3 \\ 0.4 & 0.5 & 0.3 & 0.2 & 0.1 \\ 0.1 & 0.3 & 0.2 & 0.4 & 0.5 \\ 0.5 & 0.2 & 0.1 & 0.3 & 0.4 \end{bmatrix}$$

Hidden bias = $[0.1, 0.2, 0.3, 0.1]$

$$O = W * H^T + \text{bias}$$

$$= [0.275 \ 0.425 \ 0.4 \ 0.5] * W$$

$$= [0.2 * 0.275 + 0.4 * 0.425 + 0.4 * 0.1$$

$$+ 0.5 * 0.5 + 0.1] \quad 0.66 \quad 0.3675 \quad 0.7325 \quad 0.825$$

Add hidden bias to every value.

$$= \begin{bmatrix} 0.615 & 0.66 & 0.3675 & 0.7325 & 0.825 \end{bmatrix}$$

$\downarrow$ $\downarrow$ $\downarrow$ $\downarrow$ $\downarrow$
 the cat sits on mat

SoftMax:

$$y_j = \frac{\exp(o_j)}{\sum \exp(o_j)}$$

	o_j	$\exp(o_j)$	y_j
the	0.615	1.8496	0.9643
cat	0.66	1.934	1.008
sits	0.3675	1.444	0.7529
on	0.7325	2.08	1.0845
mat	0.825	2.2819	1.1879
		<u>1.9179</u>	

The model predicts "mat"

But the true centre word is "sits"